

# Footprint Forensics: Vandalism Case Analysis

## Comparative ACE-V Examination of Crime-Scene Impression CFA1296 Against Three Reference Impressions



Partial barefoot impression (CFA1296) recovered from wet concrete at a vandalism site. Compared against three suspect reference impressions: Jo Ronaldo (CFA1297), Taylor Messi (CFA1298), Casey Mbappe (CFA1299)

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### INTRODUCTION / BACKGROUND

A partial barefoot friction-ridge impression (CFA1296) was found in wet concrete near damaged equipment at a vandalism site. Barefoot impressions were provided by 3 people: Jo Ronaldo (CFA1297), Taylor Messi (CFA1298) and Casey Mbappe (CFA1299). In Australia, plantar ridges are analyzed through ACE-V principle to assist in identification/exclusion. (7) However, reliability depends on print quality, substrate distortion, and examiner interpretation. In Australia, evidential weight is increasingly a requirement of courts, (8) and is expressed as a likelihood ratio (LR). This analysis compares CFA1296 to the three impressions and calculates an LR, then critically evaluates the results with consideration to the vandalism scenario.

### OBJECTIVE

To compare CFA1296 to three reference prints, calculate a likelihood ratio (LR), and examine plantar friction-ridge limitations

### METHOD — ACE-V

- Analysis (CFA1296):** Partial forefoot impression preserving the distal metatarsal/phalangeal area. Clear core and delta; inward-curving ridges indicate a right forefoot. 12 intervening ridges; slightly degraded but sufficient for comparison.
- Comparison:** Reference prints compared at Level 1 (pattern, geometry, ridge flow). Level 2 conducted only if Level 1 was compatible ( first 10 ridges from core).
- Exclusion:** Inconsistent Level 1 prints excluded; consistent prints advanced to Level 2 minutiae evaluation.
- Verification:** Second examiner required for operational reporting.

### ANALYSIS & RESULTS

**Laterality:** CFA1296 is a right forefoot. Core and delta both visible; ridge count between them = 12. Tracing ridges 1–10 from the core identified minutiae on ridges 1 (End), 3 (Bifurcation), 5 (End), 7 (Dot) and 9 (End); no notable minutiae on ridges 2, 4, 6, 8 or 10.

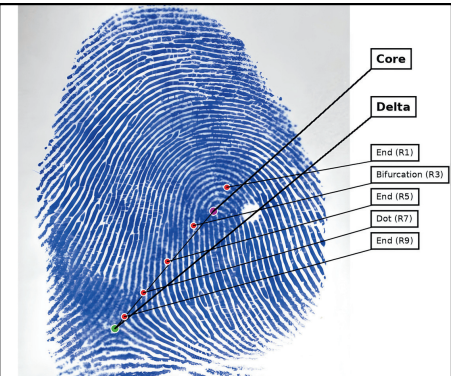


Fig. 1. CFA1296 (crime-scene impression)  
Annotated with core, delta, and Level 2 minutiae within the first 10 ridges. Chain of custody maintained.

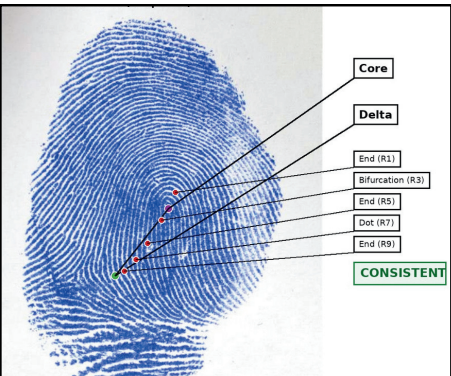


Fig. 2. CFA1296 vs CFA1297 (Ronaldo)  
Level 1 & 2 consistent;  
NOT EXCLUDED AS A SOURCE.



Fig. 3. CFA1296 vs CFA1298 (Messi)  
Annotated with core, delta, and Level 2 minutiae within the first 10 ridges. Chain of custody maintained.



Fig. 4. CFA1296 vs CFA1299 (Mbappe)  
Non-corresponding core/ridge morphology, excluded.

Ronaldo  
NOT EXCLUDED

Messi  
EXCLUDED

Mbappe  
EXCLUDED

**Likelihood Ratio Calculation:** When the frequencies are given

- Ridge ending (End): 63.2% (0.632)
- Convergence: 38.2% (0.382)
- Dot: 15.1% (0.151)

$$0.632 \times 0.632 \times 0.632 \times 0.382 \times 0.151 = 0.0145$$

**LR = 1 / 0.0145 ≈ 69** Supports Ronaldo CFA1297 over an unknown, unrelated person

### INTERPRETATION AND EVALUATION

#### A. Interpretation

Ronaldo's print was not excluded, but this does not prove he committed the vandalism. The print only shows that his impression was present at the scene. Its timing is unknown and it does not directly link him to the damage. Additional evidence such as CCTV and toolmarks is required.

#### B. Evaluation & Limitations

The combined frequency of 0.0145 (LR ≈ 69) is an estimate based on the product rule, assuming minutiae types are independent; this ignores spatial configuration, which is important to examiners. The wet concrete caused compression, distortion and loss of detail due to foot movement. (6) Plantar measurements are also affected by substrate distortion and variable pressure. (3) ACE-V remains somewhat subjective. (5) Australian examiners showed 94.8% agreement on final identifications, although disagreement increases in suitability cases. (1) Population frequencies should be interpreted cautiously because plantar minutiae vary with demographic and biomechanical factors; an Australian-specific database would strengthen the statistical basis. (4) Transparent LR reporting supports the scientific defensibility of friction-ridge evidence in Australian courts. (4)

### CONCLUSION

The first two suspects (Messi and Mbappe) were excluded at Level 1. Jo Ronaldo was not excluded as the source of CFA1296 (moderate LR ≈ 69). However, the concrete substrate caused degradation and distortion of the crime-scene impression. The simplistically calculated LR is a clear measure of the evidential weight. The physical and partial impressions and the subjective components of ACE-V, however, indicate that footprint evidence must be analyzed with reference to associative intelligence and not as evidence of identity. Separate verification and incorporation with other investigative evidence continues to be vital to a sound forensic determination.

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